CVIN Map of Work and Integration — the research side

CVIN-SC · MASc thesis · state of the research-side lanes at the freeze, 2026-10-31 · companion to the code-side branch map (05_INTEGRATION_MAP.html) · v0-3, 2026-11-09 · working document, not sent

Twelve months of written research ran as parallel lanes in a project knowledge store (168 filed documents at the freeze) and, from September, a hub repository (`nikhilprakash24/cvin-sc`, private). This map shows where each lane stands, where it enters the unified seven-chapter thesis ruled on 30 October (D-R9), and what is filed, what sits only in the hub, and what still lives only in chat. The code-side map does the same for the branches; the two maps meet in Chapter 6 and in the numbers-of-record rule.

Three things to know first

One thesis, two spines until 30 October

The hub holds a specification-first thesis (five chapters, ≈60,200 prose words, RQ1–RQ3); the code side holds a benchmark-first thesis (seven chapters, ≈45,000 words, RQ1–RQ5, H1–H6). The same labels mean different questions on the two sides. Ruling D-R9 adopted one unified outline on the hub spine with the applied system as Chapter 6 and the code side's questions as sub-questions. Nothing sent to a supervisor should present either side as "the thesis" alone.

The numbers of record are the hub's, for the interface

Interface gas per function is the Lineage A protocol matrix (N = 5, D-R8, 128/128 cells); deployment gas the N = 5 run of 2026-10-23; `createPresentation` 62,458. The code side's native-analogue gas is admissible only as *indicative, different toolchain*, in its own table. Test counts are stated per suite with date and commit (133 sandbox; 564 + 376 spoke) and never summed with the September 201.

61 output sets are unfiled

The chapter drafts of Sessions B–F, the Thesis-B trunk, the CEV ledger v2.0, the Narrative Spine, CORE 3, the closure contribution and the T-1/ISB/cost thread outputs exist in chat only (unfiled register v0-2; 21 at priority 0). Take 1 had to reconstruct ten sections by description for that reason. Filing them (D-18) is the one-hour task everything after depends on.

Lane × unified chapter

in filed or in the hub, placed in a draft of record · on produced, filed or in the hub, not yet placed · gap chat-only or not yet produced · — not applicable

Lane / thread	Ch1 Intro	Ch2 Background	Ch3 IMinimalSSI spec	Ch4 Method	Ch5 Inventory & evaluation	Ch6 Vehicular (code side)	Ch7 Discussion	Appendix
Specification & derivation G-A1, G-A2, Pass 3-Deep, §2.3 v2-0, completeness contribution	in Table 1.1 C1–C3	—	in §2.3 v2-0 promoted whole; CR-1…CR-3; partition 3/5/4	—	—	—	in §5.1 gap as structural	on M-1 universe (filed); M-2 closure table + F11 (chat only)
Evaluation of record §2.5 pre-reg, sandbox v1.1/v1.3, D-R7/D-R8, Phase 1 record, G-RIM Ch4 parts	in evidence states	—	—	in §2.5 v1-1 + deviation note; N=5 vs N=30 roles	in Ch4 §4.2–4.4 v1-2 protocol gas; coverage declared classes; gap rubric verdicts, census date	—	on concession set §5.4	on protocol gas tables; App. A/B v0-2 (filed); two Ch4 lineages to re-base (R0)
Standards & families Building Paths v4.0, Family E v2.0, Family G,	in §1.3.1 third bound	on standards-symmetry table; related-work	—	—	in Ch3 v0-9 §3.1, 3.3–3.6; gap §3.2 census (provisional only); crosswalk Table 3.0-b owed	on M-5 numbering of LSP8 / 725xy / MOBI-VID-V2	in §5.3 convergence (corroborates)	on I-1 inventory

ERC census v0-1, G-D1		audit						
MOBI VID G-MV1 ×3, WO-7/WO-V seed build, populations classification	—	on §2.4 vehicle identity (code-side §2.3 leads)	—	—	in §3.5 conformance mapping (licensed wording)	on code-side VID I/II + 5-backend sweep	on primitives-first argument	gap VID source text (D-14)
Placement thread G-P1, P2, P3 (HAA-1), P4, DPA-0…3, HDA campaign	—	—	—	gap HDA "placement as evaluation axis" collides with pre-registered §2.5 (RIM-Q14) — new slot needed	on §4.7/§4.8 placement subsections (candidate slots)	on hybrid anchoring vs code-side off-chain anchors	in §5.1 placement = partition paragraph	on HAA-1; DPA (unfiled)
Security & privacy G-C1, T-1 catalogue, ISB-1/2, B5 I2I channel	on weapon argument, sober register (one paragraph)	in §2.4 SCMS gap, tiers, K-15	—	—	on §4.6.4 revocation window (C16 proposed); §4.8.4 Ω (C17 proposed)	on merge with code-side 54-scenario suite + attestEvent fix	on limitations	on S-1 risk register; T-1 (chat only); Ω/α charter values owed (D-15)
Cost, scale, latency G-CO1 prompts 1–3, payload ladder, ISB-2, tier coverage	—	in three-tier framework (adopted, not measured)	—	—	in §4.5 invocation profile (illustration); §4.6 lower bounds; gap scale arithmetic recomputed by arm (FW-5)	on code-side scaling §5.9 (M-9, short)	on "throughput not binding" paragraph	gap Scale and Latency chats: no recoverable record
Agentic generalisation G-E1, App. I wrapper spec, G-WHY, Lane H handoff	in RQ3; C14 at Version 2	on §2.7 agent identity (merge with code-side §2.6)	—	—	—	on code-side Steward as constructive evidence (M-7, R-1 amendment dated 10-30)	in §5.2 φ mapping; gap Stage 31 switch date (Q-XD-6)	on App. I; G-E1 transfer table to be read cell by cell (D-06)
Roots G-R1 Phase 1/2, panel R9, erratum	—	on reservoirs CB-TL / CB-IG for §2.1–§2.2 (budget-neutral)	—	—	—	—	on figures S.12–S.14	gap symbol-table collision Formal Layer v2.0 vs R4 (D-23)
Vision & future programme P5 dossier, bold-claims ledger, P5C IoV, IDTX catalogue, societal integration	on one sentence	—	—	—	on ICBC Class 7→5 worked example (§4.5 / §5)	on IDTx condensed into §7.4 (M-9)	in §5.5 future work (≤ L1 claims)	gap one potential-showing test (D-19) not chosen
Governance & SSoT Scaffolding v0.4 / v2.1, Manual v2.0, Registry v13.1, sectioning, CEV, cruxes, accuracy ledger	in locked wordings and rulings govern every chapter; gap SSoT ruling (v0.4 filed vs v2.1 roll, D-06); CEV v2.0 and Narrative Spine chat-only							on ruling manifest; corrections ledger
Writing lanes Sessions B–F, Thesis-A quarry, Thesis-B trunk, Take 1	in Take 1 v0-2 continuous draft (chat + repo); hub Ch1–Ch5 drafts of record; gap Sessions B–F drafts, trunk, Thesis-A quarry unfiled (D-18)							on placeholder census (93)
Drafting programmes G-GD1 (0/10), G-SD1 (0/10), G-T1 Takes 2–4	on instruments filed; gap G-SD1 gated on the AI-use agreement (R-0); Take 3 needs the trunk pasted in							on provenance ledger, ICM, timeline (planned)

Front matter	on Abstract ≈345 words (Take 1 D-01); lay summary and preface stub in Take 1; gap GenAI preface statement (needs the origin census); title (decision 5)

Where each thing lives

Store	What	Count at freeze	State
Project knowledge store (claude.ai "CVIN Thesis")	Filed documents, v0-1/v0-2 pairs; instruments; plans; threads; reports	168 docs; 1.92 of 2.00 MB	on near cap — drafts go to chat + repo; registry rule R-F4 (retire superseded duplicates) awaits authorisation (D-1)
Hub repository `cvin-sc` `main` @ `d9f9fa3`	Sandbox v1.1 (June baseline) and v1.3 (eight options); 7 verification folders; doc corpus in 13 categories (00–12); chapter drafts; spoke import (text @839aab9, grand report @291a801)	docs 00–12; 529 SHA256SUMS entries (post-boundary)	in pushed 2026-10-20; CI runs G0 and the suites; private
Chat record (lanes)	Sessions B–F chapter drafts; Thesis-B trunk; Thesis-A quarry; CEV v2.0; Narrative Spine; CORE 1–3; T-1; ISB; cost prompts; closure contribution; Take 1 v0-1/v0-2	97 chats to 14 Oct + ≥7 lanes since; 61 unfiled sets	gap not searchable by the project; file-in owed (D-18)
Author's notebooks	March–May full pass (100+ pages); three-circle framing; 2026 update	—	gap only the author can present
Code-side repository `CVIN-SC-Implementation-SSI-DID`	Applied system, claim register, roots 27 papers, phase 2, zkp, side papers, related-work workspace, this outreach folder	23 branches; 572 commits	see the code-side map

Where the two maps meet

Join	Rule of record	Source
Pre-registration disclosure	§2.5 v1-1's dated note discloses the eight-option deviation (WO-1…9); the eleven ERC-1056 runs before the protocol are listed by a dated disclosure paragraph drafted 5 Nov (v1-2, G-P2; trace keys to confirm at source) — to be appended to the hub draft and mirrored on the code side	Phase 1 record; G-IRR §9.5
Research questions	Hub RQ1–RQ3 principal (⟦RQ-ADOPTION⟧ still owed under Q16); code-side RQ1/H1 → RQ2a, RQ3/H2 → RQ2b, RQ2/H5 → RQ2c, H4 → RQ2d, RQ4/H3 → RQ-V, RQ5/H6 → RQ-V2 + RQ3a; H1 split (create supported, update not); "all five supported" withdrawn	Merge proposal C.3; D-R9 M-2, M-6
Chapter 6	Code-side Ch4–Ch5 condensed ≈50%; native-analogue gas in its own labelled table (§6.7); never ratioed against protocol gas	C.2, C.4 rule 4; M-3
Option numbering	Canonical O1–O18 via the WO-R crosswalk; CVIN-Combined = O18 reference build, not a standard; ERC-725xy/735 and LSP8 as variants of existing O-numbers; MOBI-VID-V2 an unnumbered application profile	M-4, M-5; D-R7 c.3
ZK and the Steward	In Ch6 §6.8 and Ch7 §7.2 as the constructive rung the hub's FW-3 lacked; R-1 scope amendment dated 2026-10-30	M-7
Locked wordings on imported text	Six "first" priority claims reworded per SC-04; "validation" → corroboration; "the minimal set" never; EUDI date only in derivation form	C-9…C-13; migration step 4
Citations	One BibTeX namespace, IEEE numeric; the code side's three key systems converted	M-8; migration step 3
Freeze	Spoke thesis text frozen at tag `thesis-of-record/text-2026-10-30` (839aab9); grand report at 291a801; future spoke work lands in the hub	M-10
Known stale values on the code side	Infrastructure SPaT ratio 1.226× → 1.197×; V2V warm verify 0.153 → 0.142 ms; P* 609 → 641; ERC-4337 "second-highest" → third	code-side README; C-8

Decisions board (research side)

ID	Decision	Owner	Blocks
R-0 / AI-use	Send the AI-use disclosure and agreement brief, with the authorship timeline, before any chapter is relied on; treat the code-side documents already sent as covered by it	Candidate → supervisor	G-SD1 Stages 3–10; any draft shown
Q16 / D-R9 M-2	Adopt hub RQ1–RQ3 + sub-questions as the thesis questions	Candidate (supervisor view)	Ch1 §1.3; Ch7 §7.7; every cross-reference
R-ADM-1 / R-ADM-2	Coverage at declared classes vs rubric re-run; side-document manifest vs manifest rule	Supervisor	Ch5 verdict cells; §5.4 concessions
D-04 / RIM-Q15	Contribution count: C15 withdrawn; C16/C17 admitted?; second "C16" (HDA) folded or renumbered	Candidate	Table 1.1; abstract

D-14 (spoke) / decision 5	One title for the unified thesis	Candidate (supervisor view)	front matter
D-18	File the 61 unfiled output sets	Candidate	Take 2; Take 3 (trunk); every reconstruction in Take 1
D-1 (G-GD1/G-SD1)	Authorise registry rule R-F4 (retire superseded duplicates) to lift the store cap	Candidate	filing of G-SD1 Stage 2 onward
FW-0	Execute and date the ERC census with a second rater (κ ≥ 0.80)	Candidate	§3.2; crux X2 MED → LOW
Q-XD-2 · Q-XD-6 · D-06 · D-23 · D-15 · D-19	C11 retitled; Stage 31 switch date; SSoT roll; symbol tables; Ω/α charter values; the one potential-showing test	Candidate	as listed in the lane table
R0 (G-IRR)	Re-base the G-RIM Chapter 4 parts on the hub's §4.2–4.4 (one Ch4 lineage)	Candidate / lane	Ch5 assembly

Suggested integration order (research side)

1. Send the AI-use brief with the timeline and contribution map (R-0). Nothing else is shown until it is acknowledged.
2. File the 61 unfiled output sets (D-18); authorise R-F4 (D-1) so the store can take the G-SD1 instruments.
3. Rule the one-line decisions: Q16, D-04/RIM-Q15, Q-XD-2, Q-XD-6, D-06, D-23, title.
4. Execute and date the ERC census (FW-0); confirm the trace keys and append the §2.5 v1-2 amendment (drafted 5 Nov).
5. Re-base the two Chapter 4 lineages onto one (R0); recompute the scale arithmetic by arm (FW-5).
6. Run the D-R9 migration steps 2–4 (crosswalk, citation unification, locked-wording sweep on imported text) — these touch both sides and must precede any chapter merge.
7. Run Take 2 (panel rewrite) starting with Chapter 4 and Chapter 3; paste the trunk and run Take 3; then G-SD1 Stages 3–10 on the unified spine.
8. Cut the submission draft section by section from the best take, with origin tags carrying the record; preface GenAI statement last, from the census.

Sources: Precursor Report G-PRE v0-2; Phase 1 Consolidated Record G-P1 v0-2; Hub vs Spoke Comparison v0-1 and STATUS.md D-R9; Programme Atlas G-VIZ v0-3; Unfiled Outputs Register G-VIZ v0-2; G-SD1 Master Plan v0-2; Take 1 Panel Note G-T1 v0-2; code-side outreach folder README. Every cell is editorial placement of a recorded fact; none promotes an evidence state.